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The inclusion of vehicle shape and aerodynamic drag estimations within the life cycle energy optimisation methodology

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Abstract

The present work describes a widening of the scope of the Life Cycle Energy Optimisation (LCEO) methodology with the addition of shape-related design variables. They describe the curvature of a vehicle which impacts its aerodynamic drag and therewith its operational energy demand. Aerodynamic drag is taken into account through the estimation of the drag coefficient of the vehicle body shape using computational fluid dynamics simulations. Subsequently, the aforementioned coefficient is used to calculate the operational energy demand associated with the vehicle. The methodology is applied to the design of the roof of a simplified 2D vehicle model which is both mechanically and geometrically constrained. The roof is modelled as a sandwich structure with its design variables consisting of the material compositions of the different layers, their thicknesses as well as the shape variables. The efficacy of the LCEO methodology is displayed through its ability to deal with the arising functional conflicts while simultaneously leveraging the design benefits of the underlying functional alignments. On average, the optimisation process resulted in 2.5 times lighter and 4.5 times less life cycle energy-intensive free shape designs. This redesign process has also underlined the necessity of defining an allocation strategy for the energy necessary to overcome drag within the context of vehicle sub-system redesign.

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1. Introduction

The transport sector accounts for about one third of the total energy demand in Europe [1]. Road vehicles account for up to 82 % of that total energy demand [1], and from 3 % [2] to 50 % [3] of the energy produced by the motor is used to overcome aerodynamic drag, depending on vehicle operating conditions. This resistance that the vehicle encounters while moving through the air stems from two primary sources: skin friction drag and pressure drag [4]. The former is a direct result of the friction of the air against the “skin” of the body moving through it, while the latter mainly depends on the vehicle’s general shape [5]. This geometry dependent component of the drag constitutes up to 40 % of the total drag [5], and is particularly sensitive to the shape of the rear of the vehicle [6]. Hence, drag reduction in general, and the aerodynamic optimisation of

the rear of the vehicle in particular, continues to be one of the primary tasks of designers and aerodynamicists.

Despite the widespread use of numerical methods to estimate the drag coefficient of vehicle designs, it is still challenging to foresee the impact of aerodynamically driven design choices on the entirety of the vehicle product system. This challenge is further exacerbated by the division of the engineering design process along disciplinary boundaries. The amalgam of these two aspects further exacerbates the impact of the lack of vehicle-specific knowledge facing the designers during the early stages of the design process [7], and subsequently hampers their ability to innovatively curb the environmental impacts associated with the transport sector.

The Life Cycle Energy Optimisation (LCEO) [8] is a suggested framework, which is intended to find the optimal design variable values that minimise the life cycle energy of a product system under functional requirements. In this context, the aforementioned energy acts as a life cycle environmental proxy allowing for the inclusion of environmental considerations early-on during the design process. Furthermore, this approach avoids sub-optimal shifting of burdens between the different phases of

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the life cycle while taking into account the realities surrounding environmental design research [9].

The previous works on the LCEO methodology were published by O'Reilly et al. [8] and Jank et al. [10]. The former investigated the impact of the inclusion of the production and use energies within the LCEO framework. It was shown that the life cycle energy-optimal design of a flat sandwich panel was impacted by the consideration of the life cycle energy or the use-phase energy as the optimisation's objective function [8]. The latter presented an investigation of the inclusion of End-Of-Life (EOL) modelling into the LCEO framework. It showed that the introduction of a closed-loop recycling model into the LCEO framework affected highly the production and use energies, which, in turn, demonstrated the sensitivity of the resulting designs to the different EOL scenarios. Both works featured the life cycle energy optimisation of the design of a flat structure with a fixed associated drag coefficient.

In this work, the redesign of an aerodynamically influential vehicle sub-system under early stage design conditions is presented, with the intent of investigating the effect of the inclusion of shape-related design variables and aerodynamic drag estimations on the life cycle energy-optimal design solution. This is achieved through optimising the thickness, the material composition as well as the curvature of the roof panel of a simplified vehicle-like shape for a minimal life cycle energy. The impact of the curvature of the roof is accounted for through its effect on the resulting aerodynamic drag coefficient and its associated aerodynamic use energy, in addition to the increased geometrical stiffness.

The novelty of this work resides in the inclusion of a mechanically constrained aerodynamic shape optimisation within a wider, more complex framework, wherein the shape greatly impacts the stiffness properties of a given design, as well as its associated use energy. Within this context, the life cycle energy proxy allows for the coupling of the aerodynamic and structural properties and to balance them against the inherent energies of the different phases of the life cycle.

From a methodological standpoint, this work differs from the previous ones by the needed framework extensions to account for the aerodynamic effect, and the more robust optimisation process which relies on a hybrid optimiser. This optimiser consists in the coupling of a population based optimiser with a gradient-descent based one.

Section 2 describes the different building blocks of the used framework. Section 3, presents the optimisation results for a fixed-roof configuration as well as a free curved-roof one for a driving distance of 60,000 km. Section 4 examines the validity of the results through the vernacular of this study's enabling assumptions. In conclusion, future improvement avenues for the framework are identified.

2. Methodology

For the sake of clarity and self-containment of the present work, Sections 2.1 and 2.2 present a brief overview of the LCEO framework and the EOL modelling as introduced in [8]

and [10]. However, Sections 2.3, 2.4 and 2.5 mainly focus on the methodological developments introduced in this work. The geometry of the car model and its parametrisation are presented, and are accompanied by a summary description of the setup of the aerodynamic flow simulations that were performed in order to estimate the aerodynamic drag coefficient of a given design.

2.1. The Life Cycle Energy Optimisation Methodology

The LCEO method formally integrates environmental considerations into a design methodology through the formulation of a mathematical multidisciplinary design optimisation framework. The life cycle energy is used as the objective function to be minimised, as detailed in [8]. Functional requirements stemming from structural mechanics, and eventually other disciplinary fields, act as constraints on the design. Thus, this methodology does not compromise design requirements, but rather changes design variables so as to find the design solution which features the minimum life cycle energy use while fulfilling all the transport related functional requirements.

The life cycle energy is formulated as

$$E_L(X) = E_P(X) + E_U(X) + E_E(X) \quad (1)$$

where E_L is the life cycle energy, E_P is the production energy, E_U is the use-phase energy, E_E is the end-of-life energy, and X is the set of design variables. The production energy is obtained from the embodied energy of the constitutive materials of a given design, while the EOL energy is obtained from the energy credit or burden associated with processing the constitutive materials through different recycling or waste handling scenarios. Finally, the use phase energy is obtained through multiplying the energy required to move the vehicle according to a prescribed drive cycle (New European Drive Cycle) by the number of such cycles during the entire use phase of a vehicle. The associated optimisation problem is expressed as

$$\min(E_L(X)), \quad (2)$$

subject to constraints of the form:

$$T_{(I)}(X) \leq 0, \quad (3)$$

$$T_{(E)}(X) = 0, \quad (4)$$

$$X_{min} \leq X \leq X_{max}. \quad (5)$$

Equation 3 corresponds to functional requirements expressed as inequalities, while Equation 4 refers to the functional requirements which are expressed as equalities. The last equation, Equation 5, is the set of design variable boundaries.

2.2. End-Of-Life treatment methodologies

The EOL phase of the roof panel's life cycle has been included in the framework through the modelling of a number of recycling processes for different candidate materials, within an assumed closed-loop recycling model. In particular, the "substitution with a correction factor" method [11] has been

adopted within a closed-loop model in order to avoid the system boundary expansions inherent to open-loop models. Within this model, the recycled material is assumed to partially or completely substitute the input material; the substitution quota is accounted for through a correction factor. This modelling reflects an assumed practice of using a recycled material to produce the same product as the one from which the recycled material originates. However, depending on the employed recycling methodology, material properties can be degraded, leading to a partial substitution that is supplemented with the production of virgin material destined to the same end product.

Recyclate quality, yield and technological readiness level are the main characteristics of a recycling process. The most technologically mature approaches for the different candidate materials were selected, and used as building blocks to design four EOL scenarios, which span a wide array of technological combinations and recyclate quality levels. Table 1 shows the energy credits and burdens incurred by the processing of Carbon Fibre (CF) reinforced polymers through landfilling, pyrolysis, incineration or milling; Glass Fibre (GF) reinforced polymers through landfilling, incineration or shredding; polymer-based waste through landfilling or incineration with energy recovery. The polymers constituting the core of the panel are landfilled in the first scenario and exclusively recycled through incineration with energy recovery in the remaining EOL scenarios.

Table 1. Overview of the four different EOL treatment scenarios modelled and the EOL process energy values used in the optimisation. Adapted from [10].

#	Description	E_{EOL} [MJ/kg]				
		CF	GF	PET	PVC	PUR
1	Panel landfilling - $C_f = 0$	0	0	0	0	0
2	Panel incineration - $C_f = 0$	-32	-10	-23.8	-22.9	-24
3	CF: Pyrolysis - $C_f = 1$ GF: Shredding - $C_f = 1$	30	0.17	-23.8	-22.9	-24
4	CF: Milling - $C_f = 0.3$ GF: Shredding - $C_f = 0.3$	0.27	0.17	-23.8	-22.9	-24

2.3. The curved car roof panel: an LCEO case study

In order to widen the scope of the LCEO framework through the inclusion of aerodynamic effects, the parametrised roof of a bluff body car model is considered. The roof is assumed to have a sandwich panel structure. The flat version of the isolated flat roof design case has already served as a benchmark for previous LCEO studies [8, 10]. Here, the model is expanded to further include elements of early stage vehicle design, namely, the shape and aerodynamic drag estimations.

The base model used is a two dimensional Windsor Simple Bluff Body model, see Figure 1(a). This model has been extensively used for the purposes of basic aerodynamic research into the effects of rear-end shape, diffusers, underfloor roughness and ground simulation techniques [12]. Thus, it is an adequate candidate for the study of the inclusion of basic aero-

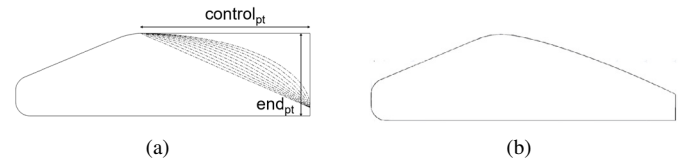


Fig. 1. (a) Illustration of the Windsor Bluff Body model featuring a parametric roof; (b) optimal aerodynamic shape of the parametrised curved roof Windsor configuration.

dynamic effects within the LCEO framework. The model is roughly the size of a quarter-scale small hatchback, measuring 1.045 m in length, 0.390 m in width and 0.290 m in height. The parametrised roof of the model is defined by the path traced by the following quadratic Bezier curve:

$$\mathbf{B}(t) = (1-t)[(1-t)\mathbf{P}_0 + t\mathbf{P}_1] + t[(1-t)\mathbf{P}_1 + t\mathbf{P}_2], \quad (6)$$

for $0 \leq t \leq 1$. $\mathbf{P}_0$ refers to the point located at the roof leading edge position, while $\mathbf{P}_1$ refers to the curve control point and $\mathbf{P}_2$ to the end-point, which is located at the intersection between the rear end and the slanted roof. The curvature of the roof is dependent on the x-coordinate of the control point and on the y-coordinate of the end-point. The former is varied between 0 and 0.6, while the latter is varied between 0 and 0.29. The full parametrisation of the model is illustrated in Figure 1(a).

The sandwich panel structural design choice introduces the following design variables: the layer thicknesses t_i , as well as the material volume fractions $V_{i,j}$. The face sheets of the panel may be composed of a blend of CF and GF as reinforcements to an epoxy matrix. The core of the panel can be composed of polyvinylchloride (PVC), polyethylene (PET) or polyurethane (PUR). The material properties of these material design variables are given in [8]. Note that the concept of hybridisation is used to allow for the continuous representation of the intra-layer material properties [13]. The Young's Moduli, densities and Poisson's ratios are proportionally determined depending on the volume fractions, $V_{i,j}$, selected.

Since one of the roof's functions within a conventional design is load carrying, two linear elastic loading responses are included as functional requirements of the design. The first load response is to a static pressure of $1 \text{ N}\cdot\text{m}^{-1}$ applied over a 33 mm long patch in the centre of the panel. The second load response is to a static pressure of $1 \text{ N}\cdot\text{m}^{-1}$ distributed over the entire top of the panel. These represent static loading cases for the roof in the event of a car roll over. The maximum displacement constraints are set to $d_{1,max} = d_{2,max} = 2.5 \times 10^{-6} \text{ m}$.

In addition to these maximum displacement functional requirements, two supplementary requirements are set for the vibrational behaviour of the panel. Two minimum frequency constraints of $f_{1,min} = 330 \text{ Hz}$ and $f_{2,min} = 520 \text{ Hz}$ are set respectively for the first and second natural frequencies of the panel. These additional requirements embody typical functional requirements stemming from flutter under driving conditions, they are also linked to the fundamental sound transmission properties of the panel.

Finally, the panel is assumed to be simply supported at the bottom two corners of the lower layer. These boundary con-

ditions (BCs) are the 2D manifestation of a plate simply supported along its edges in 3D. Thus, they are equivalent to the BCs employed in the previous LCEO benchmark studies.

The verification that these constraints are met is achieved by computing the linear elastic response and the normal modes via solving the system of elastostatic equations using the FreeFem++ partial differential equations solver [14].

2.4. Inclusion of the impact of the aerodynamic drag

The use energy is broken down into three components [15]: rolling resistance, inertial resistance to acceleration and aerodynamic drag energies. The latter is expressed as $W_D(X) = 0.5\rho_a c_D(X)C_D A(X)$, where ρ_a is the air density, $c_D(X)$ is the body height based drag coefficient of a given design, C_D is a drive cycle dependent characteristic value [15] and $A(X)$ is the frontal area of the vehicle. This area is modelled as $A(X) = (t_1 + t_2 + t_3)w + A_{Windsor}$ where t_i are the thicknesses of the different layers of the sandwich panel, w is the width of the Windsor car model and $A_{Windsor}$ is its nominal frontal area of 0.112 m^2 .

2.5. Flow Simulations

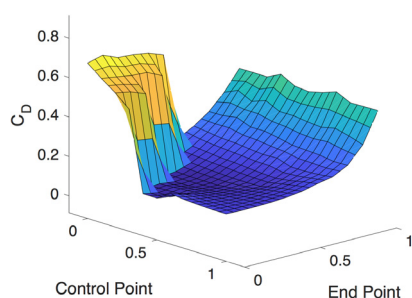


Fig. 2. Plot of the drag coefficient as a function of the control points and the end points of the quadratic Bezier curve.

The COMSOL Multiphysics Software version 5.3 is used to solve the Navier-Stokes equations for conservation of momentum and the continuity equation for conservation of mass. The steady-state Reynolds Averaged Navier-Stokes (RANS) modelling strategy [16] was adopted in this study, in conjunction with the high-Reynolds-number $k-\epsilon$ turbulence model [17] and a standard logarithmic wall function, for the prediction of the kinetic turbulent energy and eddy diffusivity. The vehicle ground clearance is set to 0.05 m . For all the computations, the free-stream velocity is equal to $20 \text{ m}\cdot\text{s}^{-1}$, which yields a body-length-based Reynolds number of 1.33×10^6 . A drag coefficient database, consisting of 121 points, and covering the range of control and end-point combinations was generated. The drag coefficient estimates were subsequently obtained through a bi-linear interpolation, and are plotted in Figure 2. This figure also shows a stark increase in drag due to complete flow separation over the rear end of the vehicle for very steep rear-end slopes. Conversely, the minimum drag coefficient, 0.106 , is obtained for a fastback-like configuration as shown in Figure 1(b).

3. Results

Table 2. Resulting optimal design variables, energies and total design masses for all four EOL treatment scenarios and for the free shape optimisation case as well as the flat roof one for the $60,000 \text{ km}$ driving distance case.

Case Scenario #	Free Roof				Flat Roof			
	1	2	3	4	1	2	3	4
$V_{1,CF} [\%]$	95	99	100	100	99	100	100	100
$V_{1,GF} [\%]$	5	1	0	0	1	0	0	0
$V_{2,CF} [\%]$	97	100	100	100	100	100	100	99
$V_{2,GF} [\%]$	3	0	0	0	0	0	0	0
$V_{c,PET} [\%]$	0	0	0	0	0	0	0	0
$V_{c,PUR} [\%]$	74	76	46	72	50	45	0	42
$V_{c,PVC} [\%]$	26	24	54	28	50	55	100	58
$t_1 [\text{mm}]$	0.13	0.13	0.52	0.14	0.45	0.47	1.37	0.52
$t_2 [\text{mm}]$	0.14	0.14	0.58	0.14	0.18	0.2	0.65	0.22
$t_c [\text{mm}]$	28	28.5	17.5	27	48	46	35	44.5
$ctrl_{pt} [\text{m}]$	0.18	0.18	0.18	0.18	0	0	0	0
$end_{pt} [\text{m}]$	0.09	0.09	0.09	0.09	0.29	0.29	0.29	0.29
$C_d [-]$	0.1	0.1	0.1	0.1	0.47	0.47	0.47	0.47
$E_P [\text{MJ}]$	46	45	117	46	97	101	218	107
$E_U [\text{MJ}]$	192	192	190	192	906	903	885	898
$E_{EOL} [\text{MJ}]$	0	-9	-111	-24	0	-23	-199	-47
$E_L [\text{MJ}]$	239	228	196	214	1004	981	903	958
$m [\text{kg}]$	0.4	0.39	0.69	0.4	0.9	0.93	1.53	0.96

The optimisation was performed for two different roof configurations, one where the variables describing the roof curvature are controlled by the optimiser, while the shape of the roof is fixed to the nominal Windsor configuration in the other. Both cases are optimised for the full set of EOL scenarios and for a $60,000 \text{ km}$ drive cycle. The resulting optimal design variables as well as their associated objective function values and total design mass are shown in Table 2. For all solutions the limiting constraint is the second vibrational functional requirement.

The design variables, which were obtained at the end of the optimisation of the curved roof case, show a high level of similarity. The face sheets are largely made up of carbon fibres, and the optimal aerodynamic shape of the parametrised Windsor car model is constant across all the EOL models. The optimal roof curvature is presented in Figure 1(b). This behaviour indicates that the overall life cycle energy is dominated by the energy needed to overcome aerodynamic drag.

Although the general trend is that the core layer presents a blend of the softer and lighter PUR with the stiffer and heavier PVC, the resulting designs show a significant amount of variation in the material compositions and the layer thicknesses of the sandwich plate across EOL models. The correction factor for the carbon fibre material in the third EOL scenario is set to one. This value makes the energy impact associated with the use of this material from the optimiser's perspective less expensive.

If the whole roof is landfilled at the end of its life cycle, a slight amount of GF is included in the face sheets, which have

a thickness of 0.13 and 0.14 mm respectively. The core layer is composed of PUR and PVC with respective material volume fractions of 74 and 26 %, and a thickness of 28 mm. Given that the EOL model does not incur any burden or credit to the system at the EOL stage, this configuration serves as a reference case.

The incineration with energy recovery of the panel introduces a non-negligible design change: GF is all but completely driven out of the final design. The heating value of CF, which is approximately three times larger than that of GF, in conjunction with its material properties, together offset its relatively energy intensive production process.

However, closing the material loop is generally preferred to incineration with energy recovery. This can be achieved from a modelling perspective through multiple approaches [11, 20]. One such approach is to assume that the CF recovered from pyrolysis, and the GF recovered from the shredding process have identical material properties to their respective virgin material counterparts. Furthermore, it is assumed that they entirely replace virgin materials in the production of new panels. These assumptions have a significant impact on the final design: it renders the access to the relatively higher performance structural properties of CF much less expensive from a life cycle energy cost perspective. This behaviour is illustrated by the four-fold increase in the CF face sheet thicknesses and the halving of the core layer thickness. The overall thickness decrease of the panel is counter intuitive from the standpoint of sandwich panel theory, since the overall stiffness of the structure is negatively impacted by a thinner core layer [18]. To compensate for this effect, it can be observed that the core layer content of the stiffer PVC is almost doubled to increase the overall stiffness of the core.

The assumption that the recycling processes enabling a closed material loop leave the material properties of the recycle intact is rather far-fetched [19]. The final EOL model does away with that assumption by considering a recycling scenario where the recycle material properties are significantly reduced. Namely, only 30 % of the recycle stemming from CF milling and GF shredding is assumed to be able to replace virgin material production. This configuration results into a final design which is largely similar to the reference case, with the only noteworthy difference being at the level of the life cycle energy. The inclusion of the EOL model results in an life cycle energy decrease from 239 to 214 MJ.

Overall, it can be noticed from Table 2 that the inclusion of EOL modelling in the LCEO framework yields heavier less life cycle energy-intensive designs. It can also be observed that PET does not figure within the core layer material mixture in the resulting designs.

The aforementioned effects of the inclusion of EOL modelling can also be observed in the case of the flat roof optimisation. However, the differences between the two case studies mainly underline the effect of the presence of curvature, or lack thereof. The effect of the functional alignment between the aerodynamic and the structural requirements is evident. The flat roof cases are on average 2.5 times heavier than their curved counterparts, and 4.5 times more energy intensive from a life cycle point of view. Furthermore, these designs are markedly

asymmetrical. The top layers are on average two times thicker than the bottom ones while the cores contain larger material volume fractions of PVC. This is especially apparent in the third EOL design case where the core layer is almost entirely made of PVC. These material compositions, which result in stiffer cores, give a quantitative representation, in terms of material composition and layer thicknesses, of the effect of the curvature of the roof.

4. Discussion

The results underline the likely gains of performing early-stage design from a holistic perspective. Optimising the system in its entirety allows for capitalizing on potentially non-trivial functional alignments. In doing so, the process could lead to non-intuitive designs which perform better from an life cycle energy standpoint in particular, and any environmental impact metric in general.

However, the allocation of credits and burdens has to be carefully performed. In fact, its effect cannot be overstated. The modelling choices in this study stem from a mass-based allocation of the energies relative to rolling-resistance and inertial resistance to acceleration [15]; while the entire energy needed to overcome the aerodynamic drag of the whole vehicle is completely attributed to the roof. Aerodynamic drag is mass independent. Thus, a mass-based allocation would not only disregard the underlying physics at the origin of the energy demand, but would also result in an increased energy demand for identically shaped roof panels with different material compositions. Contrastingly, avoiding a priori assumptions by not performing any allocation leads to a situation where the use phase energy is dominated by the aerodynamic drag energy. The design landscape is significantly impacted by this behaviour, since the optimisation of the entire system is akin to a two-step sequential one. Firstly, the shape of the design is optimised for a minimum drag coefficient, thereby minimising the aerodynamic drag energy. Secondly, The material composition of the roof panel and its thicknesses are selected by the optimiser to minimise the total life cycle energy. In other words, the potential effect of the EOL models on the shape of the design is completely obfuscated by the overwhelming shape-related energy, and leads to completely identical roof shapes regardless of the recycling methodologies included within the life cycle model.

The sensitivity of the optimal designs to the EOL models carries both upsides and downsides. On the one hand, it shows the high integration of the EOL phase of the life cycle with the two previous ones. In other words, an optimal shifting of burdens amongst the different phases of the life cycle can be reached, leading to life cycle energy-optimal designs. Such is the case for the EOL model including CF pyrolysis and GF shredding. The relatively energy intensive CF recycling processes are offset by the reduced reliance on virgin fibre production, which results in a decreased production-phase energy need. On the other hand, the correction factors in particular, and the EOL modelling strategies in general, introduce a set of highly influential variables which need to be cautiously chosen.

It should be noted that this effect is expected to exponentially increase the complexity of the model if the assumed closed-loop recycling were to be expanded to open-loop recycling within a technological whole-system approach [20].

The aerodynamic drag energy is not only sensitive to allocation approaches, it is also sensitive to the adopted turbulence model. The latter directly impacts the drag coefficient estimates, which, in turn, influence the energy needed to overcome aerodynamic drag. Despite having previously been used within the context of aerodynamic shape optimisation [6], the $k-\epsilon$ model is known to have a limited applicability when strong adverse pressure gradients are expected to exist in the flow simulation [21]. Such is the case for the Windsor Bluff Body model. In this particular case, the $k-\omega$ model [22] would yield more accurate estimates for the drag coefficient of the different car model shapes since it is better suited to deal with adverse pressure gradients. However, this limitation is not expected to incur any loss of generality with respect to the demonstrative intent of this study. Indeed, the intrinsic accuracy of the chosen computational turbulence model is not of the utmost importance; it is rather its ability to provide a consistent metric by which to rank and differentiate the multiple geometric configurations that is a key factor in enabling the life cycle energy optimisation. With this in mind, the $k-\epsilon$ turbulence model suits the purpose as it does provide a sufficient level of accuracy to judge the effect of the inclusion of aerodynamic drag within the LCEO framework.

5. Conclusion

In this paper, the effect of the inclusion of shape-related design variables and aerodynamic drag estimations within the LCEO framework was investigated. The shape of a sandwich panel roof of a Windsor car model was parametrised and optimised along with its material composition and layer thicknesses for two cases, a fixed flat roof shape and a free one. The results have successfully shown the ability of the LCEO methodology to identify and capitalise on the underlying functional alignment, existing between the structural and aerodynamic requirements, while dealing with conflictual design requirements in order to achieve life cycle energy-optimal designs. In particular, the free designs are on average 2.5 times lighter and 4.5 times less life cycle energy-intensive than their fixed shape counterparts. It has also been observed that the effect of EOL modelling on the optimal shape has been largely obfuscated by the allocation of the entirety of the energy needed to overcome aerodynamic drag to the roof. Thus, further work is needed to identify aerodynamic energy allocation strategies within the context of holistic vehicle sub-system redesign.

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